Introduction to R for MA Political Science Students

Fall 2026

Course Details:

Block Seminar with 4 Sessions
Session 1: Mon 31 August 2026, 15:00-18:00
Session 2: Tue 01 September 2026, 09:30-13:30
Session 3: Wed 02 September 2026, 15:00-18:00
Session 4: Thu 03 September 2026, 14:00-18:00
Room: A5,6. Building B, Room B 317

Instructors:

Johann-Friedrich Salzmann

Joshua Elias Schmidt Rodrigues

Course Description

This course provides an introduction to the statistical programming language R. The course material centers on the core tasks necessary to perform statistical analyses underlying quantitative research in the social sciences. No previous knowledge of R is necessary to follow this course, we will begin from zero by explaining the logic behind coding in general, then move to getting more familiar with R. The goal of this course is to get the students comfortable with using R and provide them with an introduction to key functions, packages, applied to prominent datasets in Political Science.

Upon successful completion of the course, students are capable of performing the complete workflow of a quantitative research project in R before conducting an analysis. This includes basic R programming, data import, data cleaning and wrangling, exploratory data analysis and visualization. Additionally, students will perform coding exercises on their own.

Teaching Format

We will cover four modules interactively while allowing time for you to try out the code yourselves and do exercises. The materials will be provided through a GitHub repository. There will be some homework.

Requirements

Please bring your own laptops and chargers. Download and install R, Positron Desktop, and RStudio IDE. See here & here for help. If already installed, make sure to update all (incl. packages) to their latest versions.

Content

1. The Logic of Programming and the R Environment

- Setup: Positron vs. RStudio & Github 101
- Package management
- Basic Functionality (Calculations, Vectors, Matrices, Lists)
- Variable types
- Accessing, subsetting, assigning, naming objects; indexing
- Piping

2. Data Visualization

- Base R plots vs. ggplot2
- The grammar of plots
- Scatter plots, Line plots, Bar plots, Pie plots
- Boxplots, Histogram, Density plots
- Color scales, Groups, Faceting
- Quarto Reports, Shiny Apps, Plot.ly

3. Data Manipulation

- The tidyverse – dplyr
- Loading and storing data (csv, dta, sav, xlsx)
- Ordering your data
- Transforming variables
- Missing values
- Merging data (join, bind_rows), restructuring data (pivot)
- Aggregation (summarise, mutate, sums & means of booleans)

4. R Programming

- Functions (normal vs. anonymous) & Operators
- For-Loops & Apply functions
- Purrr & rowwise, expand_grid
- If/Else & Vectorised operations

AI Usage Policy

We will have an honest conversation about AI usage for programming and data analysis. While this intro course is not graded and we are not going to set any strict rules on the matter, we advise and prepare students how to (not) work with AI during the fall semester Quantitative Methods lab that we teach, too.

Datacamp

Learning a new programming language can be overwhelming at first. For a gentle and interactive introduction to R (and also other programming languages) we created a Datacamp Classroom. Additionally to the material covered in this course, Datacamp provides numerous exercises for beginners and advanced users of R. By following the link above you are eligible to take all courses on Datacamp for six months for free. We specifically recommend the following two courses:

- Introduction to R
- Introduction to the tidyverse

Materials & Resources

- KUB Datalab: R for absolute beginners (Online Tutorial)
- R for Data Science 2e (Online Textbook)
- ggplot2: Elegant Graphics for Data analysis 3e (Online Textbook)
- Migration Guide: RStudio IDE → Positron Desktop (Positron Docs)
- Overview: tidyverse packages (tidyverse docs)
- Comparison: dplyr vs. base R (dplyr docs)
- AI assisted coding: Cognitive Surrender (Blog Article)
- Generate quizes to test your progress (Bibabo.ai)